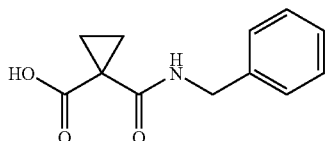


43

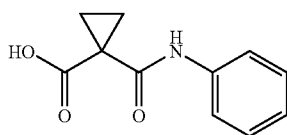
minutes at 0° C., followed by the addition of thionyl chloride (19.39 g, 0.163 mol, 1.0 eq.) with stirring for another 30 minutes at 0° C. To the resulting mixture under nitrogen was added dropwise a solution of 4-fluoroaniline (19.92 g, 0.179 mol, 1.1 eq.) in anhydrous THF (100 mL) with stirring for 1:5 hours at 0° C. The reaction mixture was diluted with ethyl acetate and washed with 1N NaOH. The layers were separated, and the ethyl acetate layer was concentrated in vacuo to give a brownish solid. The brownish solid was washed with small amount of cold ethyl acetate, filtered and dried under vacuum to yield 1-(4-fluoro-phenylcarbamoyl)-cyclopropanecarboxylic acid as a white solid (23.71 g, 65.18%). ¹H NMR (400 MHz, CD₃OD): 7.57-7.53 (m, 2H), 7.05-7.00 (m, 2H) 1.46-1.43 (m, 2H), 1.40-1.37 (m, 2H).

Example 2



1-Benzylcarbamoyl-cyclopropanecarboxylic acid. The title compound was prepared based on a modified procedure of Shih and Rankin [*Synthetic Communications*, 1996, 26(4), 833-836]: To a mixture of cyclopropane-1,1-dicarboxylic acid (5.0 g, 38.4 mmol, 1.0 eq.) in anhydrous THF (50 mL) under nitrogen was added dropwise triethylamine (3.89 g, 38.4 mmol, 1.0 eq.) with stirring for 30 minutes at 0° C., followed by the addition of thionyl chloride (4.57 g, 38.4 mmol, 1.0 eq.) with stirring for another 30 minutes at 0° C. To the resulting mixture under nitrogen was added dropwise a solution of benzylamine 5 (4.53 g, 42.3 mmol, 1.1 eq.) in anhydrous THF (25 mL) with stirring for 1.5 hours at 0° C. The reaction mixture was diluted with ethyl acetate and extracted with 2N NaOH (to pH 10). The aqueous phase was titrated with 2N HCl to pH 1-2 and then extracted with ethyl acetate. The organic phase was dried with sodium sulfate and concentrated in vacuo to give 1-Benzylcarbamoyl-cyclopropanecarboxylic acid as a white solid (4.39 g, 52.15%). ¹H NMR (400 MHz, CDCl₃): 8.44 (br s, 1H), 7.37-7.33 (m, 2H), 7.32-7.26 (m, 3H), 1.82-1.70 (m, 4H).

Example 3

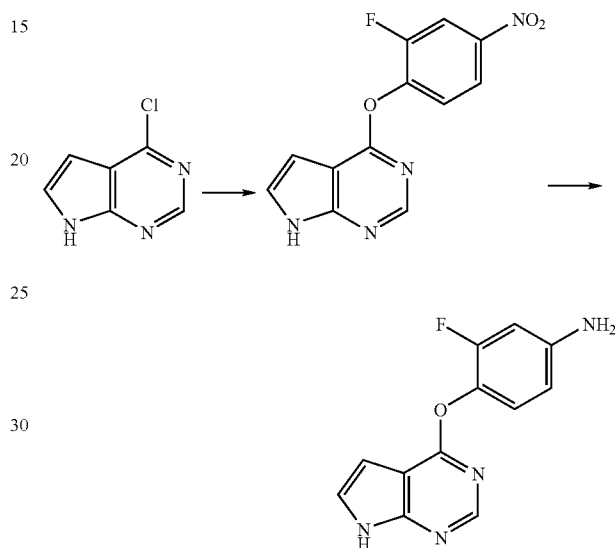


1-Phenylcarbamoyl-cyclopropanecarboxylic acid. To a mixture of cyclopropane-1,1-dicarboxylic acid (5.29 g, 40.7 mmol, 1.0 eq.) in anhydrous THF (50 mL) under nitrogen was added dropwise triethylamine (4.12 g, 40.7 mmol, 1.0 eq.) with stirring for 30 minutes at 0° C., followed by the addition of thionyl chloride (4.84 g, 40.7 mmol, 1.0 eq.) with stirring for another 30 minutes at 0° C. To the resulting mixture under nitrogen was added dropwise a solution of phenylamine 9 (4.17 g, 44.8 mmol, 1.1 eq.) in anhydrous THF (25 mL) with stirring for 1.5 hours at 0° C. The reaction mixture was diluted

44

with ethyl acetate and extracted with 2N NaOH (to pH>10). The aqueous phase was titrated with 2N HCl to pH 1-2 and then extracted with ethyl acetate. The organic phase was dried with sodium sulfate and concentrated in vacuo to give 1-phenylcarbamoyl-cyclopropanecarboxylic acid as a white solid (5.08 g, 60.8%). ¹H NMR (400 MHz, CDCl₃): 10.50 (br s, 1H), 7.56-7.54 (m, 2H), 7.35-7.31 (m, 2H), 7.15-7.10 (m, 1H), 1.94-1.91 (m, 2H), 1.82-1.79 (m, 2H).

Example 4



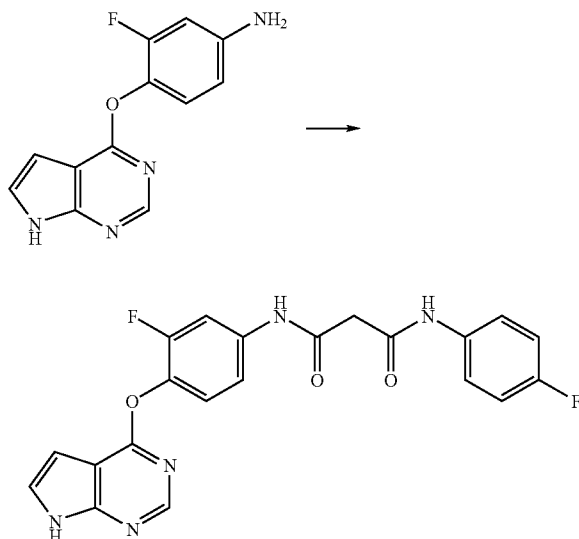
4-(2-Fluoro-4-nitro-phenoxy)-7H-pyrrolo[2,3-d]pyrimidine. To a sealed tube equipped with a magnetic stir bar was added 4-chloro-7H-pyrrolo[2,3-d]pyrimidine (1.0 g, 6.5 mmol, 1.0 eq.), 2-Fluoro-4-nitro-phenol (1.5 g, 9.5 mmol, 1.5 eq) and bromobenzene (5 ml). The mixture was stirred at 130° C. for 4 hours. The reaction mixture was then cooled to room temperature and ether was added (5 ml). After addition of ether the resulting precipitate was filtered and washed with ether. The product was recrystallized from methanol to give 4-(2-fluoro-4-nitro-phenoxy)-7H-pyrrolo[2,3-d]pyrimidine as an off white solid. (1.6 g, 89% yield). ¹H NMR (400 MHz, DMSO): 12.44 (s, 1H), 8.41 (dd, 1H), 8.34 (s, 1H), 8.24 (m, 1H), 7.80 (m, 1H), 7.61 (m, 1H), 6.69 (d, 1H); MS (ESI) for C₁₂H₇FN₄O₃: 275 (MH⁺).

3-Fluoro-4-(7H-pyrrolo[2,3-d]pyrimidin-4-yloxy)-phenylamine. To a round bottom flask equipped with a magnetic stir bar was added 4-(2-fluoro-4-nitro-phenoxy)-7H-pyrrolo[2,3-d]pyrimidine (0.5 g, 2.0 mmol, 1.0 eq.), 5% PtS (20% by weight, 120 mg) ammonium formate (451 mg, 6 eq), and ethanol (20 ml). The reaction mixture was heated at reflux with stirring for 30 min upon which the reaction was filtered hot through celite. Removal of the ethanol was followed by the addition of 1N HCl, which was further washed twice with DCM (50 ml). The aqueous layer was basified and extracted with DCM. The organic layer was then washed twice with brine (50 ml) and dried over sodium sulfate. After removal of solvent in vacuo, 3-fluoro-4-(7H-pyrrolo[2,3-d]pyrimidin-4-

45

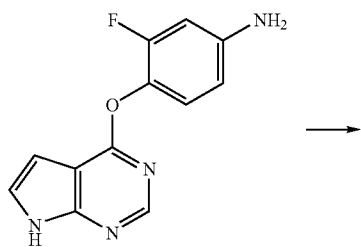
yloxy)-phenylamine as a brownish solid was obtained (0.281 g, 63%). MS (ESI) for $C_{12}H_9FN_4O$: 244 (MH^+).

Example 5



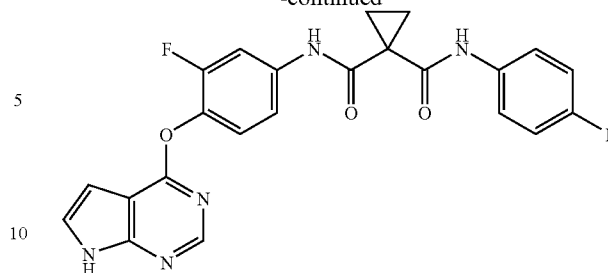
N-(4-Fluorophenyl)-N'-[3-fluoro-4-(7H-pyrrolo[2,3-d]pyrimidin-4-yloxy)phenyl]-propanediamide. To a round bottom flask equipped with a magnetic stir bar was added 3-fluoro-4-(7H-pyrrolo[2,3-d]pyrimidin-4-yloxy)-phenylamine (0.141 mg, 0.57 mmol, 1.0 eq.), 1-(4-fluorophenylcarbamoyl)-carboxylic acid (0.193 g, 0.88 mmol), HATU (0.584 g, 1.14 mmol), triethylamine (0.23 ml, 1.3 mmol) and anhydrous DMF (5 ml). The reaction mixture was stirred at room temperature overnight. The reaction mixture was diluted with H_2O (100 ml) and extracted with $CHCl_3$ (3 $\times$). The combined organic extractions were washed with H_2O (1 $\times$), sat'd $NaHCO_3$ (2 $\times$), 10% LiCl (3 $\times$), sat'd NaCl (1 $\times$), dried (Na_2SO_4), and concentrated in vacuo. The resulting crude residue was purified by HPLC to yield N-(4-fluorophenyl)-N'-[3-fluoro-4-(7H-pyrrolo[2,3-d]pyrimidin-4-yloxy)phenyl]propanediamide as a white solid (0.051 g, 21% yield). 1H NMR (400 MHz, DMSO): 12.30 (s, 1H), 10.49 (s, 1H), 10.28 (s, 1H), 8.29 (s, 1H), 7.80 (dd, 1H), 7.65 (m, 2H), 7.51 (m, 1H), 7.40 (m, 2H), 7.19 (m, 2H), 6.58 (d, 1H), 3.49 (s, 2H); MS (ESI) for $C_{21}H_{15}F_2N_5O_3$: 424 (MH^+).

Example 6



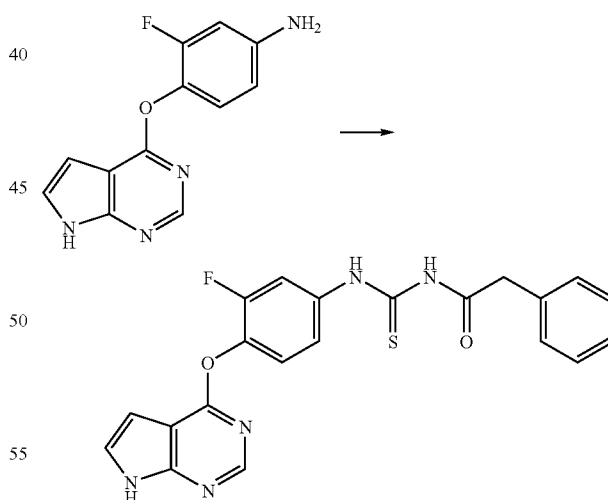
46

-continued



N-(4-fluorophenyl)-N'-[3-fluoro-4-(7H-pyrrolo[2,3-d]pyrimidin-4-yloxy)phenyl]-cyclopropanecarboxamide. To a round bottom flask equipped with a magnetic stir bar was added 3-fluoro-4-(7H-pyrrolo[2,3-d]pyrimidin-4-yloxy)-phenylamine (0.07 g, 0.29 mmol, 1.0 eq.), 1-(4-fluorophenylcarbamoyl)-cyclopropanecarboxylic acid (0.105 g, 0.32 mmol), HATU (0.304 g, 0.58 mmol), triethylamine (0.13 ml, 0.87 mmol), and anhydrous DMF (3 ml). The reaction mixture was stirred at room temperature overnight. The reaction mixture was diluted with H_2O (50 ml) and extracted with $CHCl_3$ (3 $\times$). The combined $CHCl_3$ extractions were washed with H_2O (1 $\times$), sat'd $NaHCO_3$ (2 $\times$), 10% LiCl (3 $\times$), sat'd NaCl (1 $\times$), dried (Na_2SO_4), and concentrated in vacuo. The resulting crude residue was purified by HPLC to yield N-(4-fluorophenyl)-N'-[3-fluoro-4-(7H-pyrrolo[2,3-d]pyrimidin-4-yloxy)phenyl]-cyclopropanecarboxamide as a white solid (0.011 g, 9.0% yield). 1H NMR (400 MHz, DMSO): 9.59 (bs, 1H), 8.47 (s, 1H), 7.85 (m, 2H), 7.65 (m, 3H), 7.48 (d, 1H), 7.36 (t, 1H), 7.19 (d, 1H), 7.15 (t, 3H), 1.44 (s, 5H); MS (ESI) for $C_{23}H_{17}F_2N_5O_3$: 450 (MH^+).

Example 7



N-(4-fluorophenyl)-N'-[3-fluoro-4-(7H-pyrrolo[2,3-d]pyrimidin-4-yloxy)phenyl]-propanediamide. To a round bottom flask equipped with a magnetic stir bar was added 3-fluoro-4-(7H-pyrrolo[2,3-d]pyrimidin-4-yloxy)-phenylamine (0.07 g, 0.29 mmol, 1.0 eq.), toluene (10 ml), ethanol (10 ml) and phenyl-acetyl isothiocyanate (0.22 g, 1.3 mmol, 4.5 eq). The reaction mixture was stirred at room temperature overnight. After removal of the solvent the product was precipitated from methanol to give N-(4-fluoro-4-

71

The luciferase-luciferin mix consists of 50 mM HEPES, pH 7.8, 8.5 ug/mL oxalic acid (pH 7.8), 5 (or 50) mM DTT, 0.4% Triton X-100, 0.25 mg/mL coenzyme A, 63 uM AMP, 28 ug/mL luciferin and 40,000 units of light/mL luciferase.

Structure Activity Relationships

Table 2 shows structure activity relationship data for selected compounds of the invention. Inhibition is indicated as IC₅₀ with the following key: A=IC₅₀ less than 50 nM, B=IC₅₀ greater than 50 nM, but less than 500 nM, C=IC₅₀ greater than 500 nM. Depending upon the functionality, exemplary compounds of the invention exhibit selectivity for any of c-Met, KDR, and flt-3. Abbreviations for enzymes listed in Table 2 are defined as follows: c-Met refers to hepatocyte growth factor receptor kinase; KDR refers to kinase insert domain receptor tyrosine kinase; and flt-3, fms-like tyrosine kinase-3. Empty cells in Table 2 indicate lack of data only.

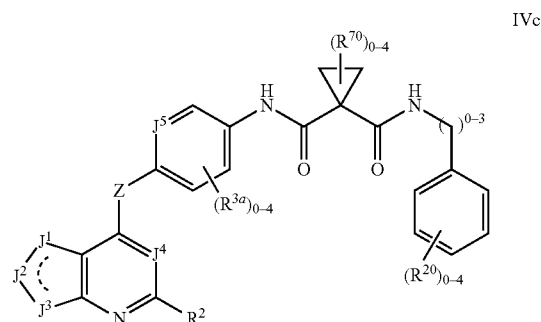
TABLE 2

Entry	Name	c-Met	KDR	flt-3
1	N-(4-fluorophenyl)-N'-[3-fluoro-4-(7H-pyrrolo[2,3-d]pyrimidin-4-yloxy)phenyl]propanediamide	B	B	A
2	N-(4-fluorophenyl)-N'-[3-fluoro-4-(7H-pyrrolo[2,3-d]pyrimidin-4-yloxy)phenyl]cyclopropane-1,1-dicarboxamide	A	A	B
3	N-([3-fluoro-4-(7H-pyrrolo[2,3-d]pyrimidin-4-yloxy)phenyl]amino)carbonothioyl]-2-phenylacetamide	A	B	C
4	N-(4-fluorophenyl)-N'-[4-{[1-(tetrahydro-2H-pyran-2-yl)-1H-pyrazolo[3,4-d]pyrimidin-4-yl]oxy}phenyl]cyclopropane-1,1-dicarboxamide	C	C	
5	2-phenyl-N-([4-{[1-(tetrahydro-2H-pyran-2-yl)-1H-pyrazolo[3,4-d]pyrimidin-4-yl]oxy}phenyl]amino)carbonothioyl]acetamide	C	C	
6	N-(4-fluorophenyl)-N'-[4-{[1-(1H-pyrazolo[3,4-d]pyrimidin-4-yloxy)phenyl]cyclopropane-1,1-dicarboxamide	C	B	
7	2-phenyl-N-([4-{[1-(1H-pyrazolo[3,4-d]pyrimidin-4-yloxy)phenyl]amino}carbonothioyl]acetamide	C	C	
8	N-(4-fluorophenyl)-N'-[4-{[9-(tetrahydro-2H-pyran-2-yl)-9H-purin-6-yl]oxy}phenyl]cyclopropane-1,1-dicarboxamide	C	C	
9	2-phenyl-N-([4-{[9-(tetrahydro-2H-pyran-2-yl)-9H-purin-6-yl]oxy}phenyl]amino)carbonothioyl]acetamide	C	C	
10	N-(4-fluorophenyl)-N'-[4-(9H-purin-6-yloxy)phenyl]cyclopropane-1,1-dicarboxamide	C	C	
11	2-phenyl-N-([4-(9H-purin-6-yloxy)phenyl]amino)carbonothioyl]acetamide	C	C	
12	N-{3-fluoro-4-[(6-{[(2-morpholin-4-ylethyl)amino]carbonyl}-7H-pyrrolo[2,3-d]pyrimidin-4-yl)oxy]phenyl}-N'-(4-fluorophenyl)cyclopropane-1,1-dicarboxamide	A	A	A

72

What is claimed is:

1. A compound of formula Vc:



wherein:

J¹ is =CH—;

J² is =C(R¹)—;

J³ is =N—;

J⁴ is =N—;

J⁵ is C—H or C—F;

Z is O;

R¹ is independently selected from halogen, trihalomethyl, —CN, —NO₂, —OR²⁰, —N(R²⁰)R²⁰, —S(O)₀₋₂R²⁰, —SO₂N(R²⁰)R²⁰, —CO₂R²⁰, —C(O)N(R²⁰)R²⁰, —N(R²⁰)SO₂R²⁰, —N(R²⁰)C(O)R²⁰, —NHCO₂R²⁰, —C(O)R²⁰, optionally substituted C₁₋₆alkyl, optionally substituted aryl, optionally substituted aryl C₁₋₆alkyl optionally substituted heterocyclyl, and optionally substituted heterocyclyl C₁₋₆alkyl;

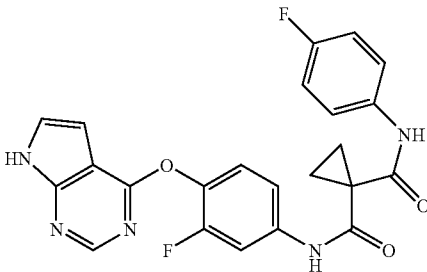
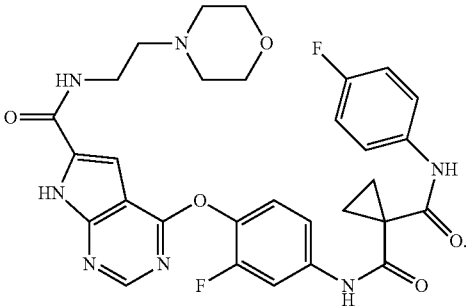
R² is H;

each R^{3a} is independently selected from —H, halogen, trihalomethyl, —CN, —NO₂, —OR²⁰, —N(R²⁰)R²⁰, —S(O)₀₋₂R²⁰, —SO₂N(R²⁰)R²⁰, —CO₂R²⁰, —C(O)N(R²⁰)R²⁰, —N(R²⁰)SO₂R²⁰, —N(R²⁰)C(O)R²⁰, —NHCO₂R²⁰, —C(O)R²⁰, optionally substituted C₁₋₆alkyl, and optionally substituted heterocyclyl;

each R²⁰ is independently —H, optionally substituted C₁₋₆alkyl, optionally substituted aryl, optionally substituted aryl C₁₋₆alkyl, optionally substituted heterocyclyl, and optionally substituted heterocyclyl C₁₋₆alkyl; or two of R²⁰, when taken together with a common nitrogen to which they are attached, can form an optionally substituted five- to seven-membered heterocyclyl, said optionally substituted five- to seven-membered heterocyclyl optionally containing at least one additional annular heteroatom selected from N, O, S, and P; and

each R⁷⁰ is independently selected from —H, halogen, trihalomethyl, —CN, —NO₂, —OR²⁰, —N(R²⁰)R²⁰, —S(O)₀₋₂R²⁰, —SO₂N(R²⁰)R²⁰, —CO₂R²⁰, —C(O)N(R²⁰)R²⁰, —N(R²⁰)SO₂R²⁰, —N(R²⁰)C(O)R²⁰, —NHCO₂R²⁰, —C(O)R²⁰, optionally substituted C₁₋₆alkyl, optionally substituted aryl, optionally substituted aryl C₁₋₆alkyl, optionally substituted heterocyclyl, and optionally substituted heterocyclyl C₁₋₆alkyl.

2. A compound selected from:

Entry	Name	Structure
2	N-(4-fluorophenyl)-N'-[3-fluoro-4-(7H-pyrrolo[2,3-d]pyrimidin-4-yloxy)phenyl]cyclopropane-1,1-dicarboxamide	
12	N-{3-fluoro-4-[(6-{[(2-morpholin-4-ylethyl)amino]carbonyl}-7H-pyrrolo[2,3-d]pyrimidin-4-yl)oxy]phenyl}-N'-(4-fluorophenyl)cyclopropane-1,1-dicarboxamide	

3. A pharmaceutical composition comprising a compound according to claim 1 or 2 and a pharmaceutically acceptable carrier.

* * * * *